The Laplace Transform Quest

From t-space to s-space and back again

APSC 2113 – Engineering Analysis I • Prof. Kartik Bulusu • Fall 2026

Name: _____ GWID: _____

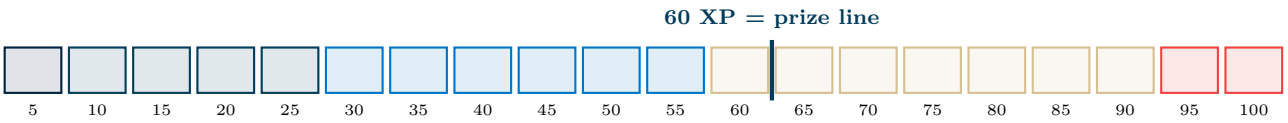
Your mission (65 minutes)

Clear four levels. Each level is a set of problems broken into steps; every correct step earns Experience Points [**XP**]. At the end of each level you compute an **unlock code** and move on. Every step tells you which entry of the **Table of Laplace Transforms** (page 3) to use: look for the tag TABLE: $f'(t)$.

The prize. Clear Levels 0–3 within 65 minutes with all four unlock codes initialled and at least **60 XP** to win the **Laplace Legend** certificate *plus* 2 bonus points on the next quiz. When you complete the final **Boss Round** you get a grade for RQ4 ad.

Level	Theme	Max XP	Target minute	Badge
0	Definition of the Laplace transform	5	10	Integral Initiate
1	Five simple transforms	20	15	Transform Apprentice
2	First-order ODE (with forcing term)	15	42	First-Order Knight
3	Second-order ODEs (with forcing term)	20	65	Second-Order Sorcerer
B	Boss Round (This is your RQ4)	40	–	Laplace Legend
	Total	**100**		

XP tracker – shade one cell per 5 XP you earn:



Pacing checkpoints

Minute 5: Level 0 done Minute 20: Level 1 done Minute 42: Level 2 done Minute 65: Level 3 done
If you are more than 5 minutes behind a checkpoint, skip to the next level's Step 1 and come back later.

Toolbox (read in 2 minutes)

1. Definition of the Laplace transform

For a function $f(t)$ defined for $t \geq 0$,

$$F(s) = \mathcal{L}\{f(t)\} = \int_0^\infty e^{-st} f(t)\, dt,$$

for every s at which the improper integral converges. It turns differentiation in t into multiplication by s, TABLE: $f'(t) \to sF(s) - f(0)$, so a linear ODE with initial conditions becomes an *algebra* problem in s.

2. The recipe for an ODE

(i) **Transform** both sides of the ODE, inserting the initial conditions.

(ii) **Solve** the resulting algebraic equation for $Y(s)$.

(iii) **Simplify** $Y(s)$ with partial fractions and/or completing the square.

(iv) **Invert** term by term, matching each piece to a line of the table.

(v) **Check** that $y(t)$ satisfies the initial conditions

3. Partial fractions fast (cover-up)

For a simple pole $s = r$ of $\dfrac{N(s)}{(s-r)Q(s)}$, the coefficient of $\dfrac{1}{s-r}$ is $\dfrac{N(r)}{Q(r)}$. This works for complex r too.

Table of Laplace Transforms

General properties of Laplace transforms and the transforms of particular functions.

Function	Transform
$f(t)$	$F(s)$
$af(t) + bg(t)$	$aF(s) + bG(s)$
$f'(t)$	$sF(s) - f(0)$
$f''(t)$	$s^2F(s) - sf(0) - f'(0)$
$f^{(n)}(t)$	$s^nF(s) - s^{n-1}f(0) - \cdots - f^{(n-1)}(0)$
$\displaystyle\int_0^t f(\tau)\,d\tau$	$\dfrac{F(s)}{s}$
$e^{at}f(t)$	$F(s-a)$
$u(t-a)f(t-a)$	$e^{-as}F(s)$
$\displaystyle\int_0^t f(\tau)g(t-\tau)\,d\tau$	$F(s)G(s)$
$tf(t)$	$-F'(s)$
$t^nf(t)$	$(-1)^nF^{(n)}(s)$
$\dfrac{f(t)}{t}$	$\displaystyle\int_s^\infty F(\sigma)\,d\sigma$
$f(t)$, period p	$\dfrac{1}{1-e^{-ps}}\displaystyle\int_0^p e^{-st}f(t)\,dt$

Function	Transform
1	$\dfrac{1}{s}$
t	$\dfrac{1}{s^2}$
t^n	$\dfrac{n!}{s^{n+1}}$
$\dfrac{1}{\sqrt{\pi t}}$	$\dfrac{1}{\sqrt{s}}$
t^a	$\dfrac{\Gamma(a+1)}{s^{a+1}}$
e^{at}	$\dfrac{1}{s-a}$
t^ne^{at}	$\dfrac{n!}{(s-a)^{n+1}}$
$\cos kt$	$\dfrac{s}{s^2+k^2}$
$\sin kt$	$\dfrac{k}{s^2+k^2}$
$\cosh kt$	$\dfrac{s}{s^2-k^2}$
$\sinh kt$	$\dfrac{k}{s^2-k^2}$
$e^{at}\cos kt$	$\dfrac{s-a}{(s-a)^2+k^2}$
$e^{at}\sin kt$	$\dfrac{k}{(s-a)^2+k^2}$
$\dfrac{1}{2k^3}(\sin kt - kt\cos kt)$	$\dfrac{1}{(s^2+k^2)^2}$
$\dfrac{t}{2k}\sin kt$	$\dfrac{s}{(s^2+k^2)^2}$
$\dfrac{1}{2k}(\sin kt + kt\cos kt)$	$\dfrac{s^2}{(s^2+k^2)^2}$
$u(t-a)$	$\dfrac{e^{-as}}{s}$
$\delta(t-a)$	e^{-as}
$(-1)^{\llbracket t/a \rrbracket}$ (square wave)	$\dfrac{1}{s}\tanh\dfrac{as}{2}$
$\llbracket\dfrac{t}{a}\rrbracket$ (staircase)	$\dfrac{e^{-as}}{s(1-e^{-as})}$

LEVEL 0 • The Definition

Badge on completion: **Integral Initiate** Target finish: **minute 5** Actual: _____

Problem 0 – Write it from memory

Step up and – State the definition **+5 XP**

Without looking at the Toolbox, write the integral that defines $F(s) = \mathcal{L}\{f(t)\}$, and state one condition on s that is needed for $\mathcal{L}\{e^{at}\}$ to exist.

Unlock code. Your value of s at which $\mathcal{L}\{e^{5t}\}$ fails to exist.

LEVEL 1 • Five Simple Transforms

Badge on completion: **Transform Apprentice** Target finish: **minute 20** Actual: _____

Problem 1.1 – $\mathcal{L}\{1\}$ from the definition

Step up and – Integrate **+4 XP**

Hint: Evaluate $\int_0^\infty e^{-st}\,dt$ as $\lim_{T\to\infty}\int_0^T$. Compare with TABLE: $1 \to 1/s$.

Problem 1.2 – $\mathcal{L}\{e^{3t}\}$ from the definition

Step up and – Integrate **+4 XP**

Hint: Combine exponentials: $e^{-st}e^{3t} = e^{-(s-3)t}$. Compare with TABLE: $e^{at} \to \dfrac{1}{s-a}$.

Problem 1.3 – Linearity

Find $\mathcal{L}\{\, 4t^2 - 3\cos 2t + 5e^{-t} \,\}$.

Step up and – Transform term by term **+4 XP**

Hint: TABLE: $af + bg \to aF + bG$, TABLE: $t^n \to n!/s^{n+1}$, TABLE: $\cos kt \to s/(s^2 + k^2)$, TABLE: e^{at}.

Problem 1.4 – The s-shift

Find $\mathcal{L}\{e^{-2t}\sin 3t\}$.

Step up and – Shift **+4 XP**

Hint: TABLE: $e^{at}f(t) \to F(s-a)$ with TABLE: $\sin kt$, or go straight to TABLE: $e^{at}\sin kt$.

Problem 1.5 – Going backwards

Find $\mathcal{L}^{-1}\left\{\dfrac{1}{s(s+1)}\right\}$.

Step up and – Split into partial fractions, and invert **+4 XP**

Hint: $\frac{1}{s(s+1)} = \frac{A}{s} + \frac{B}{s+1}$. Then use TABLE: 1 and TABLE: e^{at}.

Unlock code. Evaluate your answer to Problem 1.4 at $s = 1$.

LEVEL 2 • First-Order ODEs

Badge on completion: **First-Order Knight** Target finish: **minute 42** Actual: _____

Problem 2A – A cosine-forced first-order system (15 XP)

Solve $y' + 2y = 5\cos t, \qquad y(0) = 1.$

Step 1 – Transform both sides +3 XP

Hint: Left side: TABLE: $f'(t) \to sF(s) - f(0)$ and TABLE: $af + bg$. Right side: TABLE: $\cos kt$ with $k = 1$.

Step 2 – Solve for $Y(s)$ +2 XP

Hint: Collect the Y terms.

Step 3 – Partial fractions +4 XP

Hint: Write $\dfrac{5s}{(s+2)(s^2+1)} = \dfrac{A}{s+2} + \dfrac{Bs+C}{s^2+1}$.
Cover-up method gives A; match s^2 and constant terms for B, C.

Step 4 – Invert +3 XP

Hint: TABLE: e^{at} with $a = -2$, TABLE: $\cos kt$, TABLE: $\sin kt$.

Unlock code. [+3 XP] (Value of C Step 3) + ($\lim_{t\to 0} y(t)$ in 2A).

Scratch space

LEVEL 3 • Second-Order ODEs ✩

Badge on completion: **Second-Order Sorcerer** Target finish: **minute 65** Actual: _____

Problem 3A – Sine-forced oscillator from rest (20 XP)

Solve $y'' + 3y' + 2y = 10\sin t, \quad y(0) = 0, \quad y'(0) = 0.$

Step 1 – Transform both sides +3 XP

Hint: TABLE: $f''(t)$, TABLE: $f'(t)$, TABLE: $\sin kt$ with $k = 1$. Zero initial conditions make this quick.

Step 2 – Solve for $Y(s)$ +3 XP

Step 3 – Partial fractions (Use cover-up method at $s = -1$, $s = -2$, and $s = i$) +5 XP

Hint: $Y = \dfrac{A}{s+1} + \dfrac{B}{s+2} + \dfrac{Cs+D}{s^2+1}$. At $s = i$: $Ci + D = \dfrac{10}{(i+1)(i+2)}$; match real and imaginary parts.

Step 4 – Invert to get y(t); label the null and particular solutions +3 XP

Hint: TABLE: e^{at} twice, then TABLE: $\cos kt$, TABLE: $\sin kt$.

Step 5 – Check initial conditions +2 XP

Unlock code. [+4 XP] $A + B + C + D$ from Problem 3A, Step 3.

Scratch space

BOSS ROUND

This section is your RQ4

- Complete it, scan it and upload it on Blackboard by November 10, 2026, 11:59 PM
- +40 XP

Problem 2B – The logistic equation (non-linear!) (20 XP)

Solve $y' = y\left(1 - \frac{y}{4}\right), \qquad y(0) = 1.$

Step 1 – Linearize with $u = 1/y$ +5 XP

Hint: The table has no entry for $\mathcal{L}\{y^2\}$, so transform a *new* variable. Show $u' = -y'/y^2$, substitute, and find the linear ODE for u and $u(0)$.

Step 2 – Transform the u-equation +5 XP

Hint: TABLE: $f'(t)$ on the left; the constant $\frac{1}{4}$ uses TABLE: $1 \to 1/s$.

Step 3 – Partial fractions +5 XP

Hint: $\dfrac{1}{s(s+1)} = \dfrac{1}{s} - \dfrac{1}{s+1}$ by cover-up.

Step 4 – Invert and return to y +5 XP

Hint: TABLE: 1 and TABLE: e^{at}; then $y = 1/u$. What is $\lim_{t\to\infty} y$?

Scratch space

Problem 3B – Free (homogeneous) damped oscillator (20 XP)

Solve $y'' + 4y' + 13y = 0, \quad y(0) = 2, \quad y'(0) = -1.$

Step 1 – Transform both sides +5 XP

Hint: TABLE: $f''(t) \to s^2F - sf(0) - f'(0)$ and TABLE: $f'(t) \to sF - f(0)$.

Step 2 – Solve for $Y(s)$ +5 XP

Hint: The coefficient of Y is the characteristic polynomial $p(s)$.

Step 3 – Complete the square and split the numerator +5 XP

Hint: $s^2+4s+13 = (s+2)^2+9$. Rewrite $2s+7$ as $2(s+2)+\square$ to match TABLE: $e^{at}\cos kt$ and TABLE: $e^{at}\sin kt$.

Step 4 – Invert and check +5 XP

Hint: $a = -2,\ k = 3$. Check both $y(0)$ and $y'(0)$.

Scratch space

Score card (instructor or peer grader)

Level	Max XP	Earned	Unlock code completed?
0 – Definition	5	_____	☐
1 – Simple transforms	20	_____	☐
2 – First-order ODEs	15	_____	☐
3 – Second-order ODEs	20	_____	☐
Boss (you got RQ4 done!)	40	_____	–
Total	**100**	_____	

Laplace Legend

This certifies that

_____

cleared all four levels of the **Laplace Transform Quest**
with _____ XP,

and is awarded the **Laplace Legend** certificate *plus* 2 bonus points on the next quiz.

_____ _____

Prof. Kartik Bulusu Date

